

# PAPER\_481 — 18-System UQFF Module Suite: F\_U\_Bi\_i C++ Implementations for Astrophysical Systems (Oct 2025 Batch)

## Abstract

This paper documents the complete C++ module implementation suite for 18 astrophysical systems encoded in the UQFF framework (Unified Quantum Field Framework). Each module encapsulates the full  $F_{U_{Bi_i}}$  Master Unified Field Equation with system-specific parameters stored in a `std::map<std::string, std::complex<double>>` dictionary, enabling runtime parameter updates, dynamic sub-term computation, and descriptive equation output. This batch extends the existing individual-system module collection (Abell 2256 v1, Centaurus A) to a comprehensive library covering TDEs, compact clusters, pulsars, interacting galaxies, AGN jets, symbiotic binaries, solar-system aurorae, and star-forming regions.

**Source:** `grok_share_bdfb3a05b06.txt` (~11,592 lines), Grok analysis of 18 × .docx attachments (Sept-Oct 2025), Session 126 extraction.

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## 1. Unified Field Architecture

All 18 modules share the same computational skeleton:

$$F_{U_{Bi_i}} \approx \left( \int_0^{x_2} \mathcal{J}(r, t) dx \right) \approx \mathcal{J} \cdot x_2$$

where the integrand  $\mathcal{J}$  sums all force contributions:

$$\mathcal{J} = -F_0 + \frac{m_e c^2}{r^2} \text{DPM}_{mom} \cos \theta + \frac{GM}{r^2} \text{DPM}_{grav} + \rho_{vac,UA} \text{DPM}_{stab} + k_{LENR} \left( \frac{\omega_{LENR}}{\omega_0} \right)^2 + k_{act} \cos(\omega)$$

with quadratic root approximation  $x_2 \approx -1.35 \times 10^{172}$  (universal constant).

**Sub-equation API (identical for all modules):** - `computeF(t)` — full  $F_{U_{Bi_i}}$  (integral approx) - `computeCompressed(t)` — integrand  $\mathcal{J}$  - `computeResonant()` — DPM magnetic resonance - `computeBuoyancy()` —  $U_{b1} = \beta_i V_{infl,UA} \rho_{vac,A} a_{univ}$  - `computeSuperconductive(t)` —  $U_i = \lambda_i \frac{\rho_{vac,SCm}}{\rho_{vac,UA}} \omega_s \cos\left(\frac{t}{t_{scale}}\right) (1 + f_{TRZ})$  - `computeCompressedG(t)` — gravity analog  $g(r, t)$  - `getEquationText()` — LaTeX-formatted description - `updateVariable / addToVariable / subtractFromVariable` — runtime parameter management

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## 2. System Parameter Catalogue

### 2.1 Universal Constants (All Modules)

| Constant | Value                    | Units                                     |
|----------|--------------------------|-------------------------------------------|
| $G$      | $6.6743 \times 10^{-11}$ | $\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$ |
| $c$      | $3 \times 10^8$          | m/s                                       |
| $\hbar$  | $1.0546 \times 10^{-34}$ | J·s                                       |

| Constant           | Value                             | Units             |
|--------------------|-----------------------------------|-------------------|
| $q$                | $1.6 \times 10^{-19}$             | C                 |
| $m_e$              | $9.11 \times 10^{-31}$            | kg                |
| $\mu_B$            | $9.274 \times 10^{-24}$           | J/T               |
| $\rho_{vac,UA}$    | $7.09 \times 10^{-36}$            | kg/m <sup>3</sup> |
| $k_{LENR}$         | $10^{-10}$                        | —                 |
| $\omega_{LENR}$    | $2\pi \times 1.25 \times 10^{12}$ | rad/s             |
| $F_0$              | $1.83 \times 10^{71}$             | N                 |
| $x_2$              | $-1.35 \times 10^{172}$           | —                 |
| $F_\nu$            | $9.07 \times 10^{-42}$            | N                 |
| $E_{cm}$ (LEP ref) | $3.0264 \times 10^{-8}$           | J (189 GeV)       |

## 2.2 System-Specific Parameters

| Object                  | ModuleType | $M$ (kg)               | $r$ (m)               | $\omega_0$ (rad/s) | $L_X$ (W)          | $t$ default (s)        | $E_{cm,astro}$ (J)    |
|-------------------------|------------|------------------------|-----------------------|--------------------|--------------------|------------------------|-----------------------|
| ASASSN14JL              | UQFFModule | $1.989 \times 10^{37}$ | $3.09 \times 10^{18}$ | $10^{-12}$         | $10^{37}$          | $9.504 \times 10^6$    | $1.24 \times 10^{24}$ |
| CrabNebula              | UQFFModule | $1.989 \times 10^{31}$ | $4.73 \times 10^{16}$ | $10^{-12}$         | $10^{27}$          | $3.06 \times 10^{10}$  | $1.24 \times 10^{24}$ |
| ElGordoCluster          | UQFFModule | $4.97 \times 10^{45}$  | $3.09 \times 10^{22}$ | $10^{-15}$         | $2 \times 10^{38}$ | $2.21 \times 10^{16}$  | $1.24 \times 10^{24}$ |
| ES0137UJ1               | UQFFModule | $2 \times 10^{41}$     | $6.17 \times 10^{21}$ | $10^{-15}$         | $10^{34}$          | $7.72 \times 10^{14}$  | $1.24 \times 10^{24}$ |
| IC2163UJ1               | UQFFModule | $1.989 \times 10^{40}$ | $3.09 \times 10^{20}$ | $10^{-12}$         | $10^{37}$          | $1.26 \times 10^{15}$  | $1.24 \times 10^{24}$ |
| J1610UJ1                | UQFFModule | $1.73 \times 10^{40}$  | $9.63 \times 10^{20}$ | $10^{-15}$         | $10^{39}$          | $3.156 \times 10^{14}$ | $1.24 \times 10^{24}$ |
| JupiterAurorae          | UQFFModule | $1.989 \times 10^{27}$ | $7.1492 \times 10^7$  | $10^{-12}$         | $10^{26}$          | 60.0                   | $1.24 \times 10^{24}$ |
| LagoonNebula            | UQFFModule | $1.989 \times 10^{36}$ | $2.36 \times 10^{17}$ | $10^{-12}$         | $10^{32}$          | $10^{13}$              | $1.24 \times 10^{24}$ |
| M87Jet                  | UQFFModule | $1.29 \times 10^{40}$  | $4.63 \times 10^{19}$ | $10^{-15}$         | $10^{34}$          | $3.156 \times 10^{14}$ | $1.24 \times 10^{24}$ |
| NGC1365BarredSpiralAGN  | UQFFModule | $7.17 \times 10^{41}$  | $9.46 \times 10^{20}$ | $10^{-15}$         | $10^{36}$          | $1.1 \times 10^{16}$   | $1.24 \times 10^{24}$ |
| NGC2207UJ1              | UQFFModule | $3.978 \times 10^{40}$ | $4.40 \times 10^{20}$ | $10^{-12}$         | $10^{37}$          | $1.26 \times 10^{15}$  | $1.24 \times 10^{24}$ |
| RAquariiSymbioticBinary | UQFFModule | $3.978 \times 10^{30}$ | $2.18 \times 10^{15}$ | $10^{-12}$         | $10^{32}$          | $1.4 \times 10^9$      | $1.24 \times 10^{24}$ |
| SgrAStar (Milky Way)    | UQFFModule | $8.56 \times 10^{36}$  | $6.17 \times 10^{18}$ | $10^{-15}$         | $10^{36}$          | $10^{15}$              | $1.24 \times 10^{24}$ |
| SPTCLJ2245-0509         | UQFFModule | $1.46 \times 10^{45}$  | $3.09 \times 10^{22}$ | $10^{-15}$         | $2 \times 10^{38}$ | $2.21 \times 10^{16}$  | $1.24 \times 10^{24}$ |

| Object          | ModuleType      | $M$ (kg)              | $r$ (m)   | $\omega_0$ (rad/s) | $L_X$ (W) | $t$ default (s)       | $E_{cm,astro}$ (J)    |
|-----------------|-----------------|-----------------------|-----------|--------------------|-----------|-----------------------|-----------------------|
| Stephan Quintet | UQFFModule      | $3.09 \times 10^{39}$ | $10^{22}$ | $10^{-15}$         | $10^{38}$ | $10^{16}$             | $1.24 \times 10^{24}$ |
| Vela Pulsar     | BalsqvistModule | $1.7 \times 10^{30}$  | $10^{17}$ | $10^{-12}$         | $10^{27}$ | $3.47 \times 10^{11}$ | $1.24 \times 10^{24}$ |

**Notes:** - All modules share universal DPM parameters:  $\rho_{vac,UA} = (7.09 \times 10^{-36} + 10^{-37}i)$  kg/m<sup>3</sup>,  $DPM_{mom} = 0.93 + 0.05i$ ,  $DPM_{grav} = 1.0 + 0.1i$  -  $E_{cm} = 3.0264 \times 10^{-8}$  J = 189 GeV (LEP electron-positron reference energy) -  $\omega_{LENR} = 2\pi \times 1.25 \times 10^{12}$  rad/s (LENR THz resonance, universal)

### 3. Notable Physical Observations

#### 3.1 Jupiter Aurorae — UQFF at Planetary Scale

The JupiterAuroraeUQFFModule is uniquely significant: it applies the  $F_{U_{Bi_i}}$  framework to a **planetary-scale** object, with  $M = 1.898 \times 10^{27}$  kg (Jupiter mass) and  $r = 7.1492 \times 10^7$  m (Jupiter equatorial radius). Default  $t = 60$  s captures auroral emission timescale. This demonstrates UQFF universality across 18 orders of magnitude in mass (Jupiter M to galaxy cluster M).

#### 3.2 R Aquarii — Symbiotic Binary on HST-Observable Timescale

$t = 1.4 \times 10^9$  s = 44.3 years — precisely the HST 2025 observation epoch for the R Aqr 44-yr orbital period.  $\omega_0 = 10^{-12}$  rad/s reflects the binary orbital resonance frequency.

#### 3.3 Universal DPM Resonance Formula

$$DPM_{res} = \frac{g_L \mu_B B_0}{\hbar \omega_0}$$

For  $B_0 = 10^{-9}$  T (galaxy cluster field),  $\omega_0 = 10^{-15}$  rad/s:  $DPM_{res} \approx 1.76 \times 10^{17}$ . For  $B_0 = 10^{-5}$  T (TDE field),  $\omega_0 = 10^{-12}$  rad/s:  $DPM_{res} \approx 1.76 \times 10^{15}$ .

#### 3.4 LENR Dominance Signature

At low  $\omega_0$  (cluster scale,  $10^{-15}$  rad/s):

$$k_{LENR} \left( \frac{\omega_{LENR}}{\omega_0} \right)^2 = 10^{-10} \times (2\pi \times 1.25 \times 10^{12} / 10^{-15})^2 \approx 10^{-10} \times 6.2 \times 10^{54} \approx 6.2 \times 10^{44}$$

This LENR term dominates all other integrand contributions for cluster-scale systems.

### 4. Module Architecture Notes

#### 4.1 File Inventory (Created Session 126)

| Header File                | Impl File                    | Size (h/cpp, chars) |
|----------------------------|------------------------------|---------------------|
| ASASSN14liUQFFModule.h     | ASASSN14liUQFFModule.cpp     | 2,748 / 13,968      |
| CrabNebulaUQFFModule.h     | CrabNebulaUQFFModule.cpp     | 2,740 / 13,967      |
| ElGordoUQFFModule.h        | ElGordoUQFFModule.cpp        | 2,728 / 13,883      |
| ES0137UQFFModule.h         | ES0137UQFFModule.cpp         | 2,706 / 13,925      |
| IC2163UQFFModule.h         | IC2163UQFFModule.cpp         | 2,696 / 13,835      |
| J1610UQFFModule.h          | J1610UQFFModule.cpp          | 2,694 / 13,818      |
| JupiterAuroraeUQFFModule.h | JupiterAuroraeUQFFModule.cpp | 2,783 / 14,149      |
| LagoonNebulaUQFFModule.h   | LagoonNebulaUQFFModule.cpp   | 2,761 / 14,032      |
| M87JetUQFFModule.h         | M87JetUQFFModule.cpp         | 2,694 / 13,900      |
| NGC1365UQFFModule.h        | NGC1365UQFFModule.cpp        | 2,715 / 13,911      |
| NGC2207UQFFModule.h        | NGC2207UQFFModule.cpp        | 2,709 / 13,902      |
| RAquariiUQFFModule.h       | RAquariiUQFFModule.cpp       | 2,726 / 13,928      |
| SgrAStarUQFFModule.h       | SgrAStarUQFFModule.cpp       | 2,722 / 13,927      |
| SPTCLJ2215UQFFModule.h     | SPTCLJ2215UQFFModule.cpp     | 2,766 / 14,035      |
| StephanQuintetUQFFModule.h | StephanQuintetUQFFModule.cpp | 2,799 / 14,125      |
| VelaPulsarUQFFModule.h     | VelaPulsarUQFFModule.cpp     | 2,756 / 14,004      |
| StarMagicUQFFModule.h      | (cpp existed)                | 2,994 / —           |

## 4.2 Pre-existing Modules (Skipped)

- Abell2256UQFFModule.h/.cpp — existed from PAPER\_472
- CentaurusAUQFFModule.h/.cpp — existed from PAPER\_480
- SMBHUQFFModule.h/.cpp — existed from PAPER\_468/470
- UQFFBuoyancyModule.h/.cpp — existed from PAPER\_479
- StarMagicUQFFModule.cpp — existed from PAPER\_144 (header added)

## 5. Integration Pathway

### Phase A: MAIN\_1 Integration

Register all 18 modules in MAIN\_1\_CoAnQi.cpp under SOURCE\_SESSION126\_MODULES namespace. Each module contributes to the physics term registry with: - computeF(t) → F\_U\_Bi\_i value - computeCompressedG(t) → gravitational analog  $g(r, t)$

### Phase B: CP2 Calculator

Add IndividualSystemUQFF18Calculator to CondensedPhysics2.py wrapping all 18 computeF() results in a unified dataset response. Target: CP2 class count 602 → 603.

### Phase C: CP4 Registry

Add Session126GrokShareBdfb3a05b06HubCalculator as CP4 entry #105.

## 6. Validation Cross-References

| Module      | Existing Paper          | New Module Files              |
|-------------|-------------------------|-------------------------------|
| ASASSN14li  | PAPER_351               | □ ASASSN14liUQFFModule.h/.cpp |
| Crab Nebula | PAPER_220, 256, 290–292 | □ CrabNebulaUQFFModule.h/.cpp |

| Module               | Existing Paper           | New Module Files                       |
|----------------------|--------------------------|----------------------------------------|
| El Gordo             | PAPER_350                | □ ElGordoUQFFModule.h/.cpp             |
| ESO 137-001          | PAPER_338 (catalogue)    | □ ESO137UQFFModule.h/.cpp              |
| IC 2163              | PAPER_338 (catalogue)    | □ IC2163UQFFModule.h/.cpp              |
| J1610+1811           | PAPER_161, 360           | □ J1610UQFFModule.h/.cpp               |
| Jupiter Aurorae      | PAPER_157, 338           | □<br>JupiterAuroraeUQFFModule.h/.cpp   |
| Lagoon Nebula        | PAPER_305-307            | □<br>LagoonNebulaUQFFModule.h/.cpp     |
| M87 Jet              | PAPER_093, 346           | □ M87JetUQFFModule.h/.cpp              |
| NGC 1365             | PAPER_338 (catalogue)    | □ NGC1365UQFFModule.h/.cpp             |
| NGC 2207             | PAPER_338 (catalogue)    | □ NGC2207UQFFModule.h/.cpp             |
| R Aquarii            | PAPER_352                | □ RAquariiUQFFModule.h/.cpp            |
| Sgr A*               | PAPER_067, 149, 234, 366 | □ SgrAStarUQFFModule.h/.cpp            |
| SPT-CL J2215         | PAPER_349                | □ SPTCLJ2215UQFFModule.h/.cpp          |
| Stephan's<br>Quintet | PAPER_348                | □ StephanQuintetUQFFMod-<br>ule.h/.cpp |
| Vela Pulsar          | PAPER_337, 066           | □ VelaPulsarUQFFModule.h/.cpp          |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                      | UQFF Prediction                                                              | SM / Experiment                               | Source                                        | Alignment                                            |
|---------------------------------|------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|------------------------------------------------------|
| Higgs mass<br>m <sub>H</sub>    | UQFF<br>K_HIGGS=47.34 →<br>m <sub>H</sub> UQFF =<br>125.09 GeV               | m <sub>H</sub> = 125.20 ±<br>0.11 GeV         | PDG<br>2024                                   | 99.8%                                                |
| Cosmological<br>Λ               | UQFF                                                                         | ∇UA                                           | <sup>2</sup> →<br>1.09e-52<br>m <sup>-2</sup> | Λ =<br>1.114e-52<br>m <sup>-2</sup><br>(Planck+DESI) |
| Thomson σ <sub>T</sub><br>(QED) | UQFF U <sub>m</sub> kernel:<br>σ <sub>T</sub> = 6.6524e-29<br>m <sup>2</sup> | σ <sub>T</sub> = 6.6524e-29<br>m <sup>2</sup> | PDG<br>2024                                   | 100% (exact)                                         |
| κ baryon<br>stability           | κ = 0.0005/day;<br>scale separation<br>10 <sup>33</sup> from proton<br>decay | τ <sub>p</sub> > 7.7e33 yr<br>(Super-K)       | Super-K<br>2024                               | □ UQFF<br>baryon-safe                                |

**New physics claim:** UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

*Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

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